

Rewriting Expressions with Distributive Structure

Distributing over a sum, expanding two binomials, factoring out the greatest common factor, and checking that two forms are equivalent

Directions.

- Every answer is an *equivalent expression*, not a value of x — unless the problem asks you to substitute a number.
- Show the distribution step. Write the products before you combine.
- “Expanded form” means no parentheses left. “Factored form” means the greatest common factor is written outside one set of parentheses.
- Where a problem states units, put the unit on your final answer.

1. A mural is painted on a wall that is 6 ft tall. The mural’s width is $(2x + 5)$ ft, where x is the number of feet the artist extended it past the doorway. Write the area of the mural as an expression with no parentheses.

Answer: _____ ft²

2. A climbing gym bills each of its 9 members the same amount this month: $(4x + 15)$ dollars, where x is the number of extra classes that member took. Write the gym’s total billing, in dollars, with no parentheses.

Answer: _____

3. A carpenter cuts 6 boards to the same length, $(2x - 9)$ inches each. Write the total length of board used, in inches, with no parentheses.

Answer: _____ in

4. A rectangular tile panel has area $(18x + 24)$ square centimetres and is 6 centimetres tall. Write the area in factored form $6(\text{_____})$ so that the expression inside the parentheses is the panel's width in centimetres.

Answer: _____

5. A square patio measures x ft on each side. The owner extends it 3 ft in one direction and 8 ft in the other, so the new patio is $(x + 3)$ ft by $(x + 8)$ ft. Write the new area with no parentheses.

Answer: _____ ft^2

6. Maya claims that $4(3x + 2) + 2x$ can be rewritten as $14x + 8$. Rewrite the left-hand expression in the form $ax + b$ and state whether Maya is correct.

Answer: _____

7. A rectangle has area $(15x^2 + 10x)$ square units and width $5x$ units. Factor the area completely so that $5x$ appears outside the parentheses; the expression left inside is the rectangle's length.

Answer: _____

8. Devin wrote $5(2x - 3) = 10x - 3$. Substituting $x = 4$ into Devin's expression gives 37. Substitute $x = 4$ into the *original* expression $5(2x - 3)$ and report that value. Then state, in one sentence, what Devin did wrong.

Answer: _____

9. A rug measures $(2x + 3)$ ft by $(x + 6)$ ft. Write its area with no parentheses, in simplest form.

Answer: _____ ft^2

10. A stage crew orders 6 kits that each cost $(2x + 5)$ dollars and 3 kits that each cost $(4x - 1)$ dollars.

(a) Write the total cost $6(2x + 5) + 3(4x - 1)$ as a single expression of the form $ax + b$.

(b) Rewrite that answer in factored form, with the greatest common factor outside the parentheses.

Answer: _____